Linear Regression

MADRE: Méthodologie pour l'Analyse de Données Réseaux
(Methodology for Network Data Analysis)

Francesca Fossati

2025-2026

Outline

1. Introduction
2. Linear regression
3. Multiple linear regression
4. Potential problems

Linear Regression

Linear Regression: A very simple approach for Supervised Learning

- been around for a very long time
- predicts a quantitative response
- explains the relationship between two or more variables.

Many fancy statistical learning approaches can be seen as generalisation or extensions of linear regression.

Some history

- (1875) Sir Francis Galton originally conceived modern notions of correlation and regression, by studying eugenics of sweet pea seeds.
- (1896) Karl Pearson publishes the first rigorous treatment of correlation and regression in the Philosophical Transactions of the Royal Society of London.
- (1981) E.E. Ghiselli presents a simple proof of optimality of regression related to the sum of squared errors.

Description

Training data available: $D_n = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$
Data come in pairs $(\mathbf{x}_i, y_i)$ where:

- $\mathbf{x}_i = (x_{i,1}, \ldots, x_{i,K})$ is the input data, as a vector of size K.
- yi is the output data of size 1.

Goal: Given set of known I/O pairs, "guess" a function $f : \mathbb{R}^K \to \mathbb{R}$ that for any input $x^{(o)}$ predicts its output $y^{(o)}$.

Hopefully we get a "good" prediction. What is "good"?

Description

We are looking for an expression

$$y = f(x_1, \ldots, x_K) + \epsilon$$

which can express the output y as a deterministic function of the input $\mathbf{x}$ plus an additive random perturbation ϵ from some distribution (usually white noise $N(0; \sigma^2)$).
In this way we can do **predictions** of unknown output, given any input.
We will try to estimate the function using the existing training data.

Outline

1. Introduction
2. Linear regression
3. Multiple linear regression
4. Potential problems

Linear regression

Assumes that the output is an affine function of the input

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_K x_K + \epsilon$$

Unknowns:

- $(\beta_1, \ldots, \beta_K) \in \mathbb{R}^K$ is a vector of parameters/coefficients.
- β_0 is the bias parameter or intercept.
- ϵ is a random error term; independent of x with mean 0.

Simple linear regression

Let us simplify for just 1-dimension, $x \in \mathbb{R}$.

$$y = \beta_1 x + \beta_0 + \epsilon$$

- β_1 is the slope (average increase in y for unit increase of x).
- β_0 is the intercept term (expected value of y when $x = 0$),
- ϵ is the error term, which summarises what we miss by using a simple linear model, e.g. non-linearities, other variables, or measurement errors.

Simple linear regression

Simple Goal: Use training data to produce estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for the model coefficients. Then we can make the prediction $\hat{y}_o$ for y_o, on the basis of input x_o:

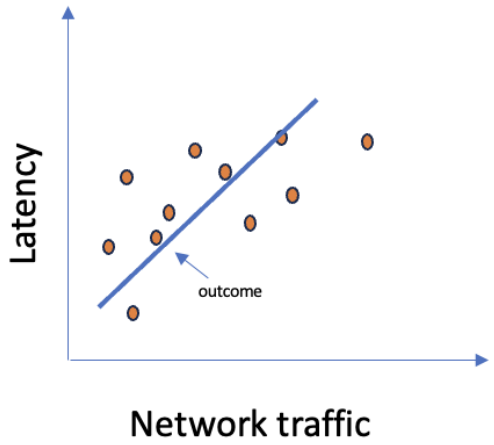
$$\hat{y} = \hat{\beta}_1 x + \hat{\beta}_0$$

Secondary Goal: Draw a line in the x-y plane that best "fits" our data points. This is called the regression line (without the error term).

Example

A network administrator wants to understand how incoming traffic affects network latency for a company server.

Data: Network traffic (Mbps) and Latency (ms)



$$\hat{y} = 12 + 0.8x$$

Interpretation:

- For each 1 Mbps increase in traffic, latency increases by approximately 0.8 ms.
- Even with 0 traffic, the base latency is 12 ms, likely due to infrastructure delays.

Residuals

The residuals are defined as:

$$\epsilon_i := y_i - \hat{y}_i = y_i - \hat{\beta}_1 x_i - \hat{\beta}_0$$

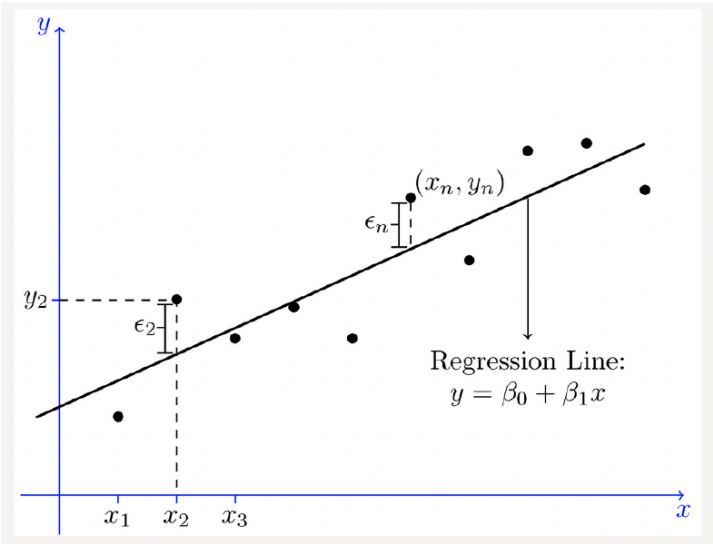


Figure: Regression line and residuals. [1]

[1] *C. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.*

RSS

The Residual Sum of Squares (RSS) is a function of $\hat{\beta}_0$ and $\hat{\beta}_1$,

$$RSS(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

It is always non-negative $RSS \geq 0$.

RSS

The Residual Sum of Squares (RSS) is a function of $\hat{\beta}_0$ and $\hat{\beta}_1$,

$$RSS(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

It is always non-negative $RSS \geq 0$.

Find and use the coefficients that minimize the RSS! This should provide a good fit to the data.

Least squares coefficient estimates

The optimal least squares coefficient estimates are found by:

$$\min \; RSS(\hat{\beta}_0, \hat{\beta}_1).$$

Solution:

$$\frac{\partial RSS}{\partial \hat{\beta}_0} = 2(-1)\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0,$$

$$\frac{\partial RSS}{\partial \hat{\beta}_1} = 2(-1)\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)x_i = 0.$$

Least squares coefficient estimates

Given $n \geq 3$ observations $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, we estimate β_0 and β_1 as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \tag{1}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \tag{2}$$

where $\bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i$ and $\bar{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$ are the sample means.

Alternative method

Consider again the linear model with random errors. Error ϵ has zero mean $\mathbb{E}[\epsilon] = 0$ and is independent of x.

$$y = \beta_1 x + \beta_0 + \epsilon$$

Applying expectation on both sides, we get

$$\begin{aligned}\mathbb{E}[y] &= \beta_1 \mathbb{E}[x] + \beta_0 + \mathbb{E}[\epsilon] \\ &= \beta_1 \mathbb{E}[x] + \beta_0.\end{aligned}$$

Alternative method

We take covariance between x and $y = \beta_0 + \beta_1 x + \epsilon$.

$$\begin{aligned}
\mathrm{Cov}(x, y) &:= \mathbb{E}\left[(x - \mathbb{E}[x])(y - \mathbb{E}[y])\right] \\
&= \mathbb{E}\left[(x - \mathbb{E}[x])(\beta_0 + \beta_1 x + \epsilon - \beta_0 - \beta_1 \mathbb{E}[x] - \mathbb{E}[\epsilon])\right] \\
&= \mathbb{E}\left[(x - \mathbb{E}[x])(\epsilon + \beta_1 (x - \mathbb{E}[x]))\right] \\
&= \beta_1 \mathbb{E}\left[(x - \mathbb{E}[x])^2\right] =: \beta_1 \cdot \mathrm{Var}(x)
\end{aligned}$$

Alternative method

$$\beta_1 = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)}\ , \qquad \beta_0 = \mathbb{E}[y] - \beta_1 \mathbb{E}[x].$$

With the observed pairs (x_1, y_1), ..., (x_n, y_n) we get the estimates $\hat{\beta}_0$, $\hat{\beta}_1$,

$$\mathbb{E}[x] \approx \frac{1}{n}\sum_{i=1}^{n} x_i := \bar{x},$$

$$\mathbb{E}[y] \approx \frac{1}{n}\sum_{i=1}^{n} y_i := \bar{y},$$

$$\mathrm{Var}(x) \approx \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 := s_{xx},$$

$$\mathrm{Cov}(x,y) \approx \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) := s_{xy}$$

Remark: For (s_{xx}) and (s_{xy}), division by (n) or $(n-1)$ does not affect the coefficients. But the same choice should be applied to both. estimators.

Exercice - Numerical example

Consider a data set of n = 4 known (x, y) samples

$$D = \{(1,3),(2,4),(3,8),(4,9)\}.$$

Find:

- The coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ for the simple linear regression.
- The residuals ϵ_i from the estimated values.

Accurancy of the model

- Mean Squared Error (MSE) and Residual Sum of Squares (RSS).
- Residual Standard Error (RSE).
- R^2 statistic or Coefficient of determination.
- Confidence intervals.
- p-value.

Mean Squared Error (MSE) and Residual Sum of Squares (RSS)

The Residual Sum of Squares (RSS) is defined as:

$$\text{RSS} := \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

The Mean Squared Error (MSE) is defined as:

$$\text{MSE} := \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n}RSS$$

- RSS tells you how well your model fits the data.
- A lower RSS means a better fit.
- MSE is a normalized version of RSS, making it easier to compare across models or datasets.

Residual Standard Error (RSE)

The residual standard error (RSE) provides the average amount that the response y deviates form the true regression line $\beta_1 x + \beta_0$.

$$\text{RSE} = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

It is a measure of lack of fit for the model, having also the nice interpretation: $Var(\epsilon) \approx RSE^2$.

R^2 statistic

Alternative name: Coefficient of determination
$R^2 \in [0, 1]$ is the proportion of variability in y (dependent) that can be explained / predicted by knowing the variability of x (independent).
The closer to 1, the better the fit.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} \tag{3}$$

Where:

- $\text{RSS} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ is the **Residual Sum of Squares**, representing the unexplained variation by the model.
- $\text{TSS} = \sum_{i=1}^{n}(y_i - \bar{y})^2$ is the **Total Sum of Squares**, representing the total variation in the observed data.

R^2 statistic

R^2 can be directly linked to the covariance between x and y.

$$R^2 = \frac{\mathrm{Cov}(x,y)^2}{\mathrm{Var}(x)\cdot\mathrm{Var}(y)} \approx \frac{s_{xy}^2}{s_{xx}s_{yy}} =: \rho^2$$

where ρ^2 is the sample correlation coefficient.
To better understand, let us take a look at the variance of $y = \beta_0 + \beta_1 x + \epsilon$,

$$Var(y) \overset{indep.x,\epsilon}{=} \beta_1^2 Var(x) + Var(\epsilon)$$

- Variance due to variation of x: $\beta_1^2 Var(x)$
- Variance of error: $Var(\epsilon)$

If $Var(\epsilon)$ is small, then y will be close to $\beta_0 + \beta_1 x$, so that our linear regression model will successfully estimate y.

R^2 statistic

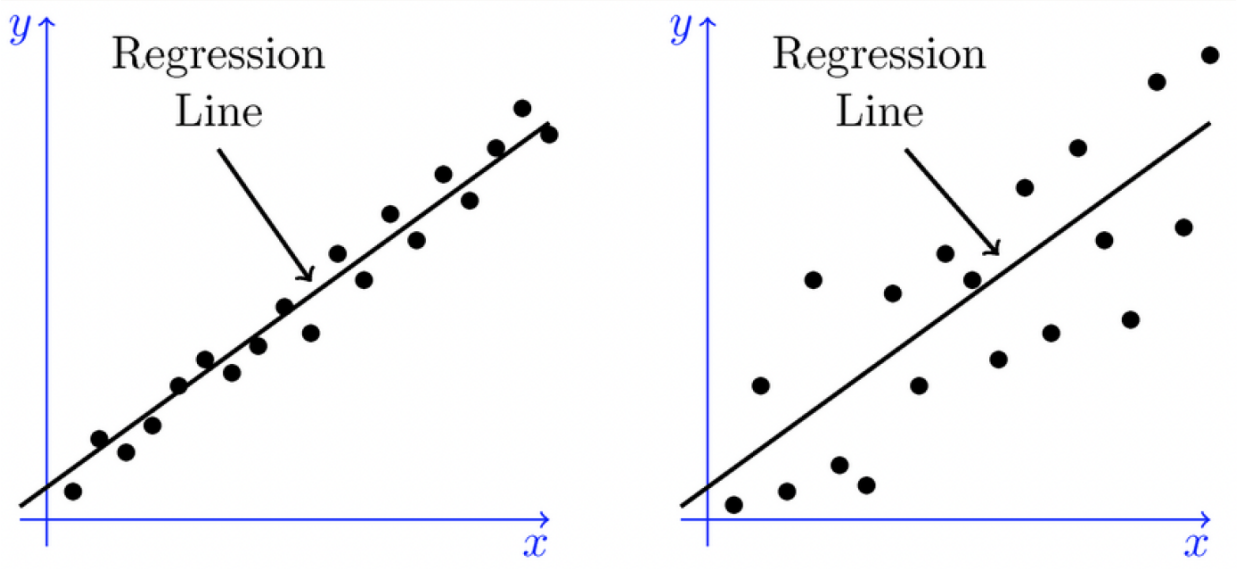


Figure: Left: high value of R^2, right: low value of R^2.[2]

[2] C. Bishop, "Pattern Recognition and Machine Learning", Springer 2006.

Confidence intervals

A confidence interval is a range of values, derived from sample data, that is likely to contain the true value of an unknown population parameter (like a mean, proportion, or **regression coefficient**), with a certain level of confidence.

The 95% connfidence intervals for β_0 and β_1 are

$$\hat{\beta}_0 \pm 2 \cdot SE(\hat{\beta}_0) \text{ and } \hat{\beta}_1 \pm 2 \cdot SE(\hat{\beta}_1)$$

where SE is the Standard Error.

$$SE(\hat{\beta}_0) = Var(\epsilon)\left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right]$$

$$SE(\hat{\beta}_1) = \frac{Var(\epsilon)}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

To estimate $Var(\epsilon)$ we use the Residual Standard Error (RSE)

$$Var(\epsilon) \approx RSE^2.$$

p-value

A p-value in linear regression is used to test whether a regression coefficient is significantly different from zero — that is, whether the predictor has a statistically significant relationship with the response.

Hypotesis test:

$$H_0 : \beta_1 = 0$$

versus

$$H_1 : \beta_1 \neq 0$$

If $\beta_1 = 0$ then the model reduces to $Y = \beta_0 + \epsilon$ and x is not associated with y. We compute the p-value. The p-value is the lowest significance level that results in rejecting the null hypothesis.

This relates to the probability that $\hat{\beta}_1$ is close to 0, related to the standard error.

If the p-value is very small ($1\% - 5\%$) we reject the null hypothesis, i.e. a relationship does exist between x and y.

Exercice - Numerical example

Consider a previous execice.
Find the:

- RSS
- MSE
- RSE
- R^2
- s_{yy}

Did we provide a good fit?

Outline

1. Introduction
2. Linear regression
3. Multiple linear regression
4. Potential problems

Multiple linear regression

Very often, a response depends on several ($K > 1$) features.

Naive approach: Run one simple linear regression per feature.
But! In this way the estimates ignore all other features left outside:
Not always good, due to correlation among features.
Correct approach: Extend the simple linear model to accommodate multiple predictors. Give to each predictor a separate slope coefficient in a single model,

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \epsilon.$$

Coefficient estimation

Again, the regression coefficients are unknown and must be estimated.
We use the formula:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

We choose the parameters that minimise again the RSS.

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)$$

Complicated expressions, better use existing software packages.

Least-squares plane

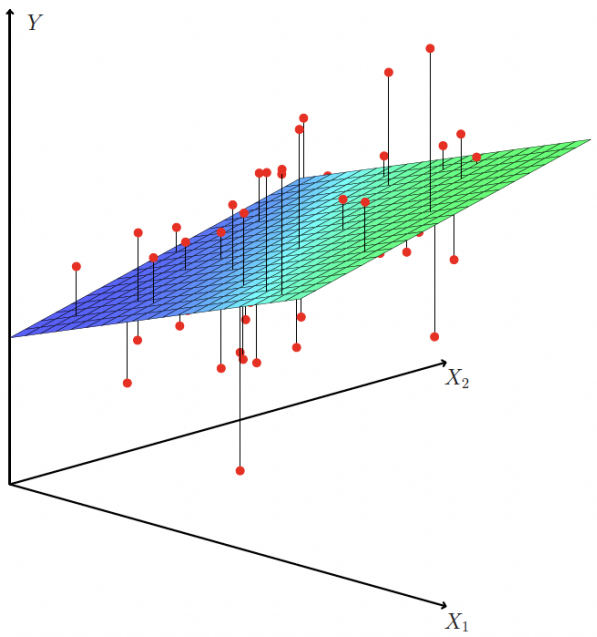


Figure: Regression plane for 2 features[3]

[3] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

Hypothesis test

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

versus

$$H_a : \text{at least one } \beta_j \text{ non-zero.}$$

To answer, we compute the F-statistic:

$$F = \frac{(TSS - RSS)/k}{RSS/(n-k-1)}$$

where: $RSS = \sum_{i=1}^{n}(y_i - \hat{y}_i)$ and $TSS = \sum_{i=1}^{n}(y_i - \bar{y}_i)$ is the Total Sum of Squares.
If the F-statistic is very close to 1 then we expect no relationship between the response and the predictors.
On the other hand, if H_a is true, we expect $F > 1$.

Variable selection

How do we decide on the important variables that influence the y? We use the F-statistic and the individual p-value.

- Use different combinations of features to derive the F-statistic. If for a specific combination among these the value of the statistic drops considerably, then this is an indicator that among the included features, some are unrelated to the response.
- From the individual p-values, the one with the highest p-value is a candidate to be removed from the model.
- The R^2 value determines how much of the data variance is explained by the model. If it is low then the feature set used does not contain much info.
- Other selection methods (forward and backward selection...)

Outline

1. Introduction
2. Linear regression
3. Multiple linear regression
4. Potential problems

Potential problems

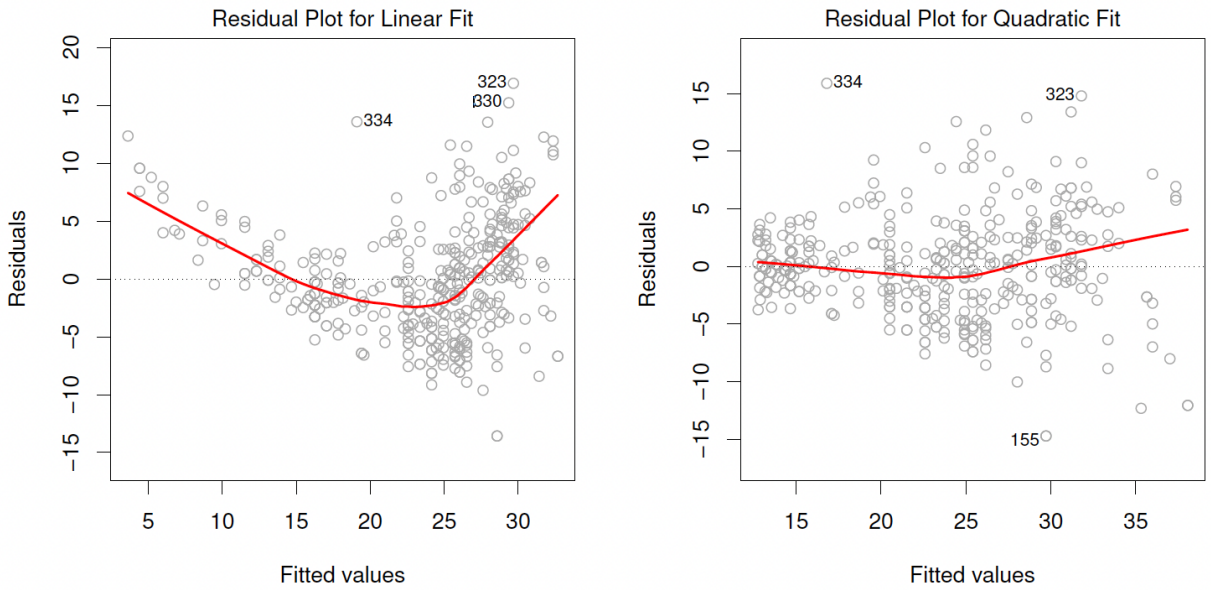
- Non-linearity of the responce-predictor relationships.
- Correlation of error terms.
- Non-constant variance of error terms (heteroscedasticity)
- Outliers (e.g. incorrect data collection)
- High leverage points.
- Collinearity.

We use the **Residual plots** to detect problems:
$(y_i - \hat{y}_i, x_i)$ or $(y_i - \hat{y}_i, \hat{y}_i)$.

Next figures are from *Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. An introduction to statistical learning: with applications in R". Springer Texts in Statistics.*

Non-linearity

Linear regression assumes a linear relationship between predictors and the response variable. If the true relationship is nonlinear, the model will miss important patterns.

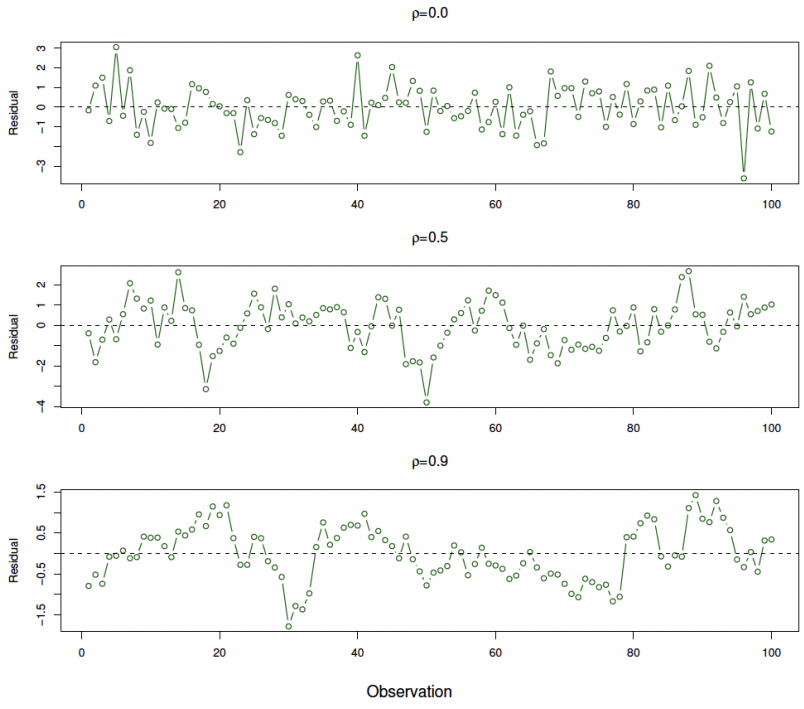


Solutions:

- Include polynomial terms
- Use nonlinear models or transformations
- Try generalized additive models (GAMs)

Error correlation

Regression assumes that the residuals (errors) are independent. If they're correlated, like in time series data, this assumption is violated.

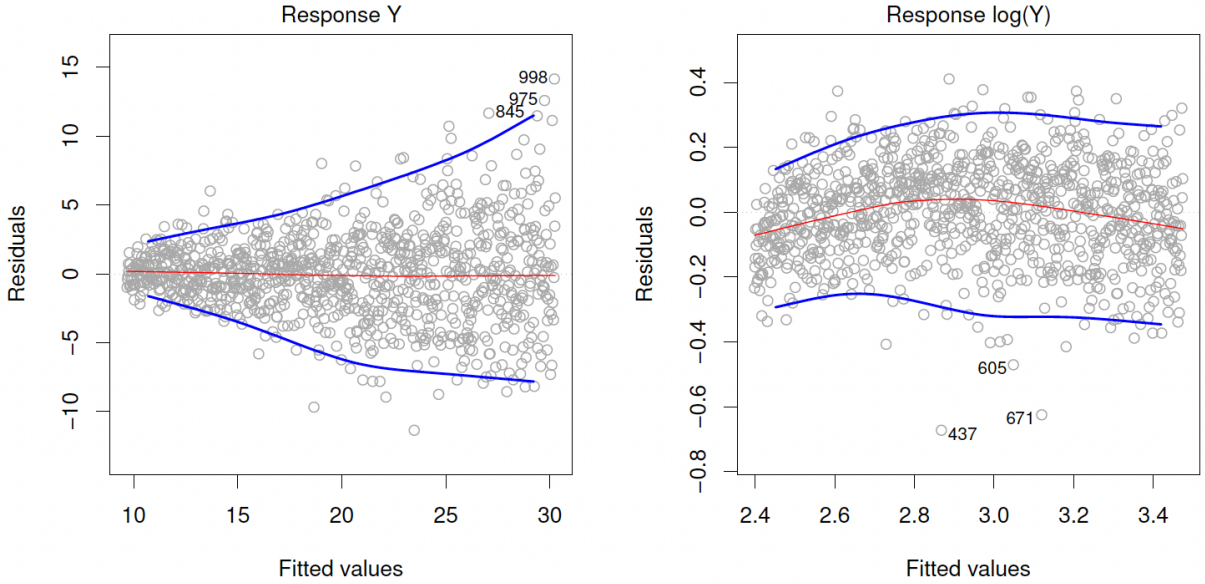


Solutions:

- Use time series models
- Include lagged variables

Heteroscedasticity

Linear regression assumes homoscedasticity — constant variance of residuals across all levels of the predictor(s). If variance changes (e.g., spreads out as X increases), it's heteroscedastic.

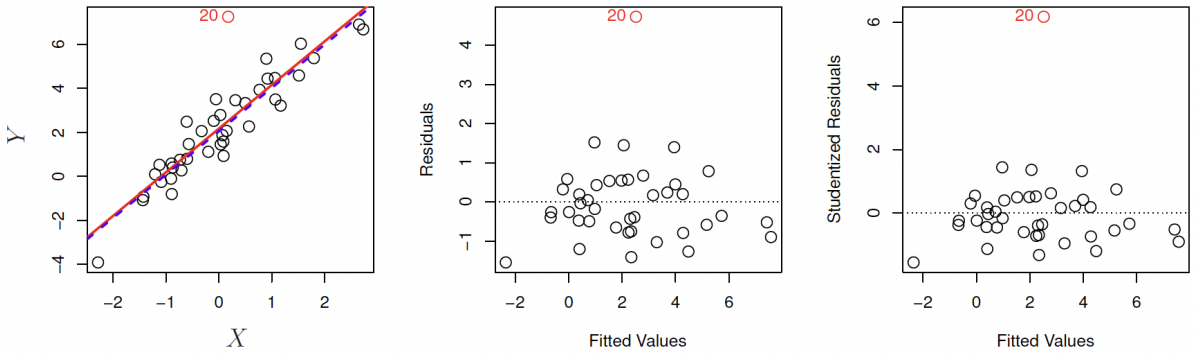


Solutions:

- Use White's test or Breusch-Pagan test to detect
- Apply transformations (e.g., log of response)

Otliers

An outlier is a data point with a response value that is far from the rest. It may be due to measurement error, data entry error, or rare events.

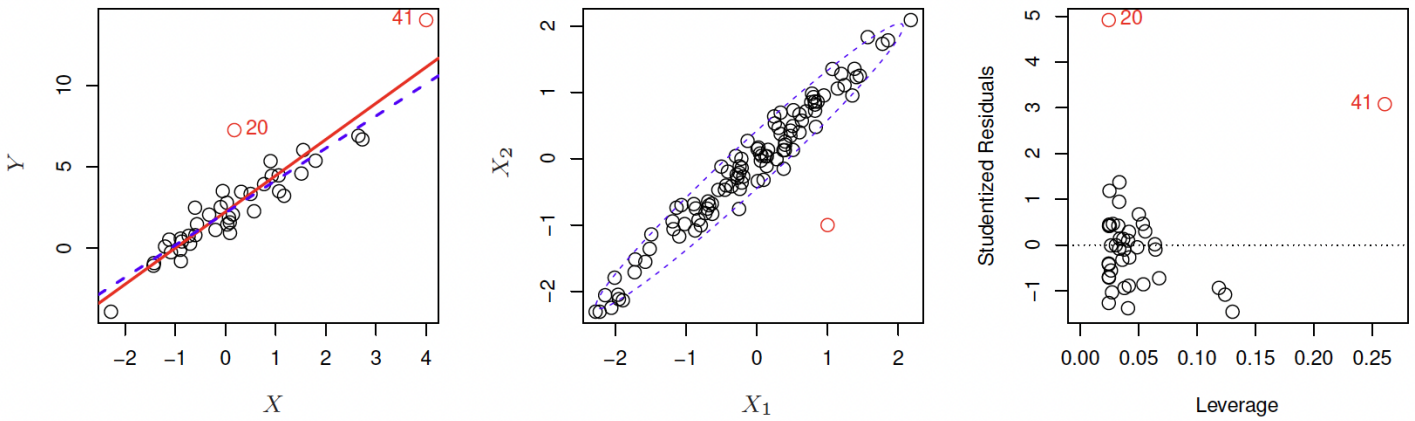


Solutions:

- Identify with residual plots, standardized residuals
- Remove or correct if data is clearly wrong
- Use robust regression if you can't remove

High leverage points

These are observations with extreme predictor values, far from the mean of X. They may or may not be outliers in Y, but they can pull the regression line toward themselves.

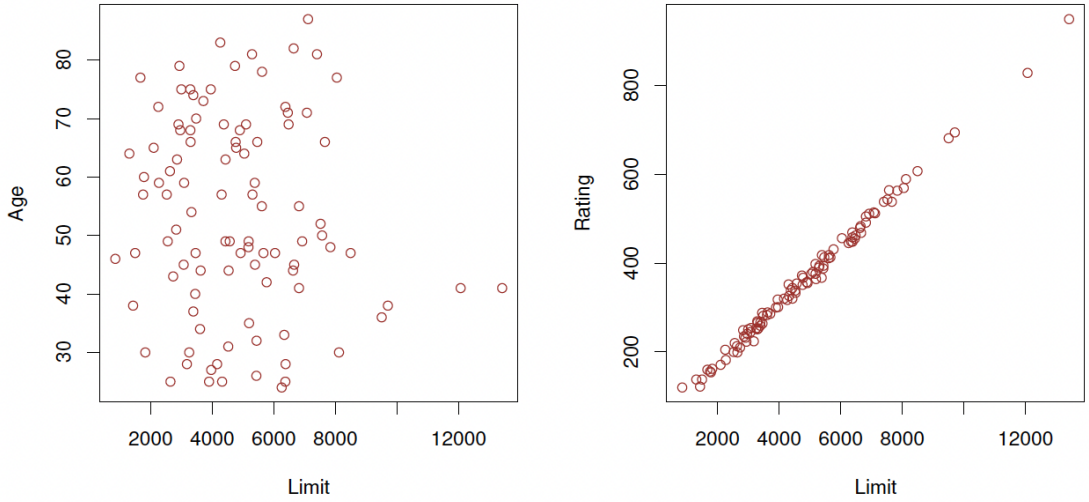


Solutions:

- Investigate these points
- Consider removing or modeling separately
- Use robust regression methods

Collinearity

Collinearity refers to the situation in which two or more predictor variables are closely related to one another.



Solutions:

- Remove one of the correlated predictors
- Combine correlated predictors